



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	⟨U er J'⟩ (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
a ⁷ S ₃ - y ⁷ P ₂ ^o	357.9708	357.8686	837.4775	5.565	1.520e+08	24.11	100	strong UV resonance line a7S3 - y7P4
a ⁷ S ₃ - y ⁷ P ₃ ^o	359.4511	359.3485	834.0285	4.922	1.510e+08	24.11	99.7	strong UV resonance line a7S3 - y7P3
a ⁷ S ₃ - y ⁷ P ₂ ^o	360.6357	360.5329	831.2888	4.181	1.510e+08	24.11	99.7	strong UV resonance line a7S3 - y7P2
a ⁷ S ₃ - z ⁷ P ₂ ^o	425.5534	425.4336	704.4768	3.361	3.300e+07	5.253	100	main cooling line: Zeeman slower, MOT, imaging
a ⁷ S ₃ - z ⁷ P ₃ ^o	427.6000	427.4797	701.1049	2.926	3.170e+07	5.053	99.9	optical pumping to m _J = -3 and "depumper" towards 5S2 /...
a ⁷ S ₃ - z ⁷ P ₂ ^o	429.0924	428.9717	698.6664	2.458	3.100e+07	4.943	99.8	429.1 nm line
a ⁵ S ₂ - z ⁷ P ₃ ^o	633.1842	633.0091	473.4680	0.160	2.900e+04	5.053	0.0914	633.2 nm repumper 5S2 -> 7P3
a ⁵ D ₂ - z ⁷ P ₃ ^o	646.8765	646.6978	463.4462	≈0.030	9.647e+02	5.053	0.00304	646.9 nm repumper 5D2 -> 7P3
a ⁵ D ₃ - z ⁷ P ₂ ^o	649.1976	649.0182	461.7892	0.007	4.200e+01	5.253	0.000127	intercombination leak of the cooling line into metastable...
a ⁵ D ₃ - z ⁷ P ₃ ^o	653.9727	653.7921	458.4174	≈0.063	4.168e+03	5.053	0.0131	654.0 nm repumper 5D3 -> 7P3
a ⁵ D ₄ - z ⁷ P ₂ ^o	658.2741	658.0923	455.4219	0.013	1.270e+02	5.253	0.000385	intercombination leak of the cooling line into metastable...
a ⁵ D ₄ - z ⁷ P ₃ ^o	663.1842	663.0011	452.0500	0.078	6.000e+03	5.053	0.0189	663.2 nm repumper 5D4 -> 7P3

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

⁵³Cr hyperfine structure ($I = 3/2$)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
a ⁷ S ₃	-83.5990(20)	-	F=3/2: +501.59 F=5/2: +292.60 F=7/2: -0.00 F=9/2: -376.20
a ⁵ S ₂	-173.820	-	F=1/2: +782.19 F=3/2: +521.46 F=5/2: +86.91 F=7/2: -521.46
a ⁵ D ₃	-35.683(5)	15.565(5)	F=3/2: +223.44 F=5/2: +122.56 F=7/2: -7.78 F=9/2: -156.68
a ⁵ D ₄	-48.755(5)	63.021(10)	F=5/2: +396.61 F=7/2: +182.64 F=9/2: -51.95 F=11/2: -276.77
z ⁷ P ₃ ^o	-1.5(20)	-2.000	F=3/2: +7.80 F=5/2: +5.55 F=7/2: +1.00 F=9/2: -7.25
z ⁷ P ₂ ^o	-11.8(4)	-0.6(6)	F=5/2: +88.20 F=7/2: +47.32 F=9/2: -5.64 F=11/2: -70.95

Measured A, B constants; sources in the data file.

Isotope shifts

Transition	λ vac (nm)	Isotopes	Shift (MHz)
a ⁷ S ₃ - z ⁷ P ₂ ^o	425.553	53-52	10.7(17)
a ⁷ S ₃ - z ⁷ P ₃ ^o	427.600	53-52	18(23)
a ⁵ S ₂ - z ⁷ P ₃ ^o	633.184	53-52	-155(53)
a ⁵ D ₃ - z ⁷ P ₃ ^o	653.973	53-52	670(28)
a ⁵ D ₄ - z ⁷ P ₃ ^o	663.184	53-52	819(28)

Shift = ν(first isotope) - ν(second isotope).